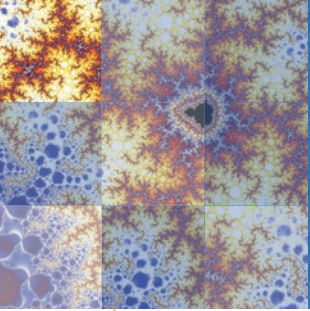




Fractional Exponents

Some radical expressions can be rewritten using fractional exponents, also known as rational exponent form.

Radicals and Rational Exponent Form

Property

For any positive integer n and any real number b ,

$$\sqrt[n]{b} = b^{\frac{1}{n}}$$

except when $b < 0$ and n is even.

Properties of Radicals and Rational Exponents

For any positive integers m and n and any real number b ,

$$\sqrt[n]{b^m} = (\sqrt[n]{b})^m = b^{\frac{m}{n}}$$

except when $b < 0$ and n is even.

EXAMPLE 1

Write the expression in rational exponent form. Assume all variables are positive.

A $\sqrt[3]{5}$

SOLUTION

$$\sqrt[3]{5} = 5^{\frac{1}{3}}$$

B $(\sqrt{x+1})^3$

SOLUTION

$$(\sqrt{x+1})^3 = \sqrt[2]{(x+1)^3} = (x+1)^{\frac{3}{2}} \quad \blacksquare$$

EXAMPLE 2

Write the expression in radical form.

A $2^{\frac{3}{4}}$

SOLUTION

$$2^{\frac{3}{4}} = \sqrt[4]{2^3} = \sqrt[4]{8}$$

B $(4m)^{\frac{2}{3}}$

SOLUTION

$$(4m)^{\frac{2}{3}} = \sqrt[3]{(4m)^2} = \sqrt[3]{16m^2} \quad \blacksquare$$

► **Remember Properties of Exponents** If a and b are nonzero real numbers and m and n are integers, then

$$a^m \cdot a^n = a^{(m+n)} \quad \text{Product of Powers}$$

$$\frac{a^m}{a^n} = a^{(m-n)} \quad \text{Quotient of Powers}$$

$$(ab)^n = a^n b^n \quad \text{Power of a Product}$$

$$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n} \quad \text{Power of a Quotient}$$

$$(a^m)^n = a^{mn} \quad \text{Power of a Power}$$

Simplifying a Radical Expression

The properties of exponents given for integer exponents may also be applied to fractional exponents, but be careful with even roots.

Properties of n th Roots of n th Powers

For any real number b and positive integer n ,

$$\text{If } n \text{ is even, then } \sqrt[n]{b^n} = |b|.$$

$$\text{If } n \text{ is odd, then } \sqrt[n]{b^n} = b.$$

Product and Quotient Properties of Radicals

For any real numbers a and b , where the following roots are represented by real numbers, the following properties hold true.

Product Property of Radicals

$$\sqrt[n]{ab} = \sqrt[n]{a} \sqrt[n]{b}$$

Quotient Property of Radicals

$$\sqrt[n]{\frac{a}{b}} = \frac{\sqrt[n]{a}}{\sqrt[n]{b}}, (b \neq 0)$$

EXAMPLE 3

Simplify the expression.

A $\sqrt[4]{(-16)^4}$

SOLUTION

$$\sqrt[4]{(-16)^4} = |-16| \quad \text{Simplify.}$$

$$= 16 \quad \text{Simplify.}$$

B $\sqrt[3]{128}$

SOLUTION

$$\begin{aligned}\sqrt[3]{128} &= \sqrt[3]{64 \cdot 2} && \text{Factor 128.} \\ &= \sqrt[3]{64} \cdot \sqrt[3]{2} && \text{Product Property of Radicals} \\ &= 4\sqrt[3]{2} && \text{Simplify.}\end{aligned}$$

C $\sqrt[3]{2^6}$

SOLUTION

$$\begin{aligned}\sqrt[3]{2^6} &= \sqrt[3]{(2^2)^3} && \text{Power of a Power Property} \\ &= 2^2 && \text{Simplify.} \\ &= 4 && \text{Simplify.}\end{aligned}$$

D $\sqrt[3]{\frac{27}{125}}$

SOLUTION

$$\begin{aligned}\sqrt[3]{\frac{27}{125}} &= \frac{\sqrt[3]{27}}{\sqrt[3]{125}} && \text{Quotient Property of Radicals} \\ &= \frac{3}{5} && \text{Simplify.} \blacksquare\end{aligned}$$

You can use properties of radicals or fractional exponents to simplify expressions.

EXAMPLE 4

Simplify the expression. Assume all variables are positive.

A $(-32)^{\frac{3}{5}}$

SOLUTION

$$(-32)^{\frac{3}{5}} = (\sqrt[5]{-32})^3 = (-2)^3 = -8$$

B $(x^5)^{\frac{1}{4}} \cdot x^{\frac{3}{4}}$

SOLUTION

$$(x^5)^{\frac{1}{4}} \cdot x^{\frac{3}{4}} = x^{\frac{5}{4}} \cdot x^{\frac{3}{4}} = x^{\frac{8}{4}} = x^2 \blacksquare$$

EXAMPLE 5

Write the expression in simplified radical form. Assume all variables are positive.

A $\sqrt[4]{32x^{10}y^4}$

SOLUTION

$$\sqrt[4]{32x^{10}y^4} = \sqrt[4]{2^5x^{10}y^4} \quad \text{Write 32 as a power of 2.}$$

$$= \sqrt[4]{2^4x^8y^4} \cdot \sqrt[4]{2x^2} \quad \text{Product Property of Radicals}$$

$$= 2x^2y^4\sqrt[4]{2x^2} \quad \text{Simplify.}$$

B $\sqrt[8]{a^6b^2}$

SOLUTION

$$\sqrt[8]{a^6b^2} = (a^6b^2)^{\frac{1}{8}} \quad \text{Property of Radicals}$$

$$= a^{\frac{6}{8}}b^{\frac{2}{8}} \quad \text{Power of a Product}$$

$$= a^{\frac{3}{4}}b^{\frac{1}{4}} \quad \text{Simplify.}$$

$$= \sqrt[4]{a^3b} \quad \text{Property of Radicals} \quad \blacksquare$$